University of Glasgow

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TITLE: Evaluating the Market Efficiency of NFTs Using Quantitative Techniques

STUDENT ID:

DATE:

WORDCOUNT:

NAME OF SUPERVISOR:

Abstract

In this dissertation, the dynamics of the NFTs market is explored using a cross-sectional study of 100 leading projects that involved variables in the floor price (in USD), total trade amount, daily amount, and owner amount. Preprocessing was done to correct the inaccuracy such as standardization of currency, labeling of variables and generating categorical tiers in creating a comparative testing. Through SPSS, the research conducted descriptive statistics, one-way ANOVA, Pearson correlation, multiple regression, and K-Means clustering to answer three research questions on the correlation about five variables, including valuation, liquidity, ownership, and market segmentation. The findings were that there were no significant variations in the trading volume across at the different price levels and that there was essentially no correlation between the floor prices and other prices in the market. Regression analysis showed inability of classical indicators to predict price variation and only 1.6 percent of the variance was explained. The hypothesis that it is possible to form meaningful groups where NFTs are concerned by performing clustering was confirmed. The results contradict the idea that NFTs have a direct positive association with higher valuation and the significance of qualitative variables, including the community involvement, branding, and rarity in terms of the influence on NFT prices. These factors create an informed picture of the behaviour of the NFT markets and serve as good strategic indicators to the investors and analysts.

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Chapter 1 - Introduction

1.1 Background: The Emergence of Non-Fungible Tokens (NFTs)

It is a digital asset that represents token ownership over a digital or physical work called Non-Fungible Tokens (NFTs), and is unique and cryptographically secured. Working on the blockchain technology, each NFT has individuated data that renders it unique and non-interchangeable, and thus the aspect of non-fungibility (Fairfield, 2021). This is in sharp contrast to fungal items such as cryptocurrency (e.g., Bitcoin) or more traditional fiat currency, in which each unit is the same and can be exchanged with any other. This special asset tokenization provides the possibility of ownership and genuineness of assets in the digital media, as can be verified.

However, the NFT market saw an explosive growth in popularity in the past several years, as what was once a niche idea has become a major component of the digital-finance ecosystem (Bruçaj and Lawrence, 2024). It has been propelled by headline sales, including the multi-million-dollar auction of digital artworks, and is increasingly crossing over into other industries, including gaming, collectibles, and virtual real estate.

1.2 Research Rationale and Problem Statement

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	A	B	C	D	E	F	G	H	I	J	K	L	M
1	floor_usd	total_volur	name	chains	floor_eth	owners_co	total_volume_eth	category	daily_volume_usd	daily_volume_eth	total_volume_usd_converted	daily_volume_usd_converted	price_tier
2	0	2.73E+09	CryptoPunk	ethereum	0	3313	775319.6491	collections	1636712	464.06	2734509359	1636713.857	Low
3	232425	1.06E+09	Bored Ape	ethereum	65.9	6088	301049.3346	collections	6313610	1790.1099	1061784290	6313618.235	High
4	12400	8.05E+08	Decentrala	ethereum	3.515817	6261	228251.7252	collections	1822724	516.8002996	805031162.8	1822725.965	High
5	51141	6.25E+08	Mutant Ap	ethereum	14.5	11199	177077.4351	collections	8617518	2443.34119	624542282.7	8617528.729	High
6	12274	4.62E+08	The Sandbo	ethereum	3.48	17884	130882.1537	collections	3671598	1041.015168	461614089.8	3671602.703	High
7	0	3.47E+08	Rarible	ethereum	0	199006	98355.78772	collections	52085	14.7677	346895402.8	52084.85803	Low
8	13579	2.8E+08	Meebits	ethereum	3.85	5495	79477.65493	collections	357593	101.389	280313276.5	357593.3741	High
9	6	2.73E+08	Decentrala	ethereum	0.001685	10210	77401.75671	collections	1597	0.4526738	272991698.8	1596.555361	Low
10	0	2.68E+08	CryptoKitti	ethereum	0	111855	70173.99924	collections	5068	1.328055665	247499799.4	4683.9786	Low
11	6525	2.55E+08	Loot (for Ad	ethereum	1.85	2523	72411.17661	collections	88343	25.048	255390199.8	88342.9054	Medium
12	34917	2.38E+08	Cool Cats N	ethereum	9.9	5045	67418.5497	collections	2017265	571.959	237781481.9	2017267.639	High
13	846	2.21E+08	SuperRare	ethereum	0.24	5855	62590.62058	collections	296598	84.095	220753643.9	296598.3963	Low
14	12168	1.95E+08	CrypToadz	ethereum	3.45	3667	55202.03792	collections	522490	148.14245	194694523.1	522490.1966	High
15	19575	1.93E+08	CLONE X -	ethereum	5.55	7410	54769.70446	collections	2281936	647.0017878	193169707	2281939.386	High
16	35	1.85E+08	Parallel Alp	ethereum	0.01	30737	52524.80116	collections	93406	26.48368	185252057.6	93406.46905	Low
17	74	1.77E+08	Sorare	ethereum	0.021	18903	50246.23881	collections	0	0	177215694.7	0	Low
18	19751	1.74E+08	Bored Ape	ethereum	5.6	5236	49344.36603	collections	1009477	286.2189442	174034839.5	1009478.326	High
19	30928	1.7E+08	Doodles	ethereum	8.769	5561	48253.53203	collections	11237779	3186.268719	170187528.6	11237792.88	High
20	988	1.7E+08	PUNKS Con	ethereum	0.28	7076	48123.79062	collections	453438	128.5642133	169729937.8	453438.8427	Low

nft_collections_with_price_tier

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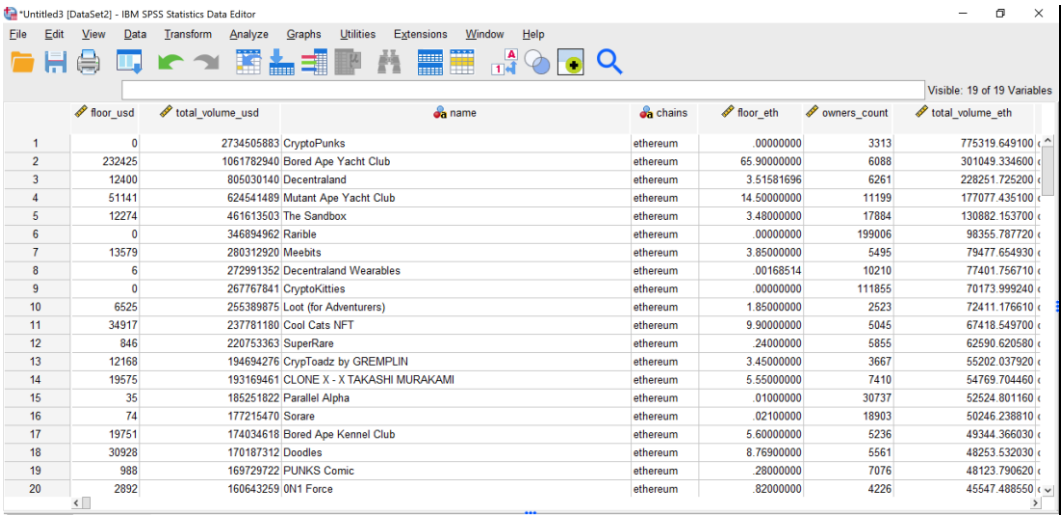

*Untitled3 [DataSet2] - IBM SPSS Statistics Data Editor

File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help

Visible: 19 of 19 Variables

	floor_usd	total_volume_usd	name	chains	floor_eth	owners_count	total_volume_eth
1	0	2734505883	CryptoPunks	ethereum	.00000000	3313	775319.649100
2	232425	1061782940	Bored Ape Yacht Club	ethereum	65.90000000	6088	301049.334600
3	12400	805030140	Decentraland	ethereum	3.51581696	6261	228251.725200
4	51141	624541489	Mutant Ape Yacht Club	ethereum	14.50000000	11199	177077.435100
5	12274	461613503	The Sandbox	ethereum	3.48000000	17884	130882.153700
6	0	346894962	Rarible	ethereum	.00000000	199006	98355.787720
7	13579	280312920	Meebits	ethereum	3.85000000	5495	79477.654930
8	6	272991352	Decentraland Wearables	ethereum	.00168514	10210	77401.756710
9	0	267767841	CryptoKitties	ethereum	.00000000	111855	70173.999240
10	6525	255389875	Loot (for Adventurers)	ethereum	1.85000000	2523	72411.176610
11	34917	237781180	Cool Cats NFT	ethereum	9.90000000	5045	67418.549700
12	846	220753363	SuperRare	ethereum	.24000000	5855	62590.620580
13	12168	194694276	CrypToadz by GREMPLIN	ethereum	3.45000000	3667	55202.037920
14	19575	193169461	CLONE X - X TAKASHI MURAKAMI	ethereum	5.55000000	7410	54769.704460
15	35	185251822	Parallel Alpha	ethereum	.01000000	30737	52524.801160
16	74	177215470	Sorare	ethereum	.02100000	18903	50246.238810
17	19751	174034618	Bored Ape Kennel Club	ethereum	5.60000000	5236	49344.366030
18	30928	170187312	Doodles	ethereum	8.76900000	5561	48253.532030
19	988	169729722	PUNKS Comic	ethereum	.28000000	7076	48123.790620
20	2892	160643259	0N1 Force	ethereum	.82000000	4226	45547.488550

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```
DESCRIPTIVES VARIABLES=floor_usd total_volume_usd daily_volume_usd owners_count
  /SAVE
  /STATISTICS=MEAN STDDEV MIN MAX.
```

Descriptives

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
floor_usd	100	0	617215	25529.55	88098.319
total_volume_usd	100	30650206	2734505883	140037334.4	302610477.7
daily_volume_usd	100	0	11237779	758365.55	1688545.717
owners_count	100	0	219997	10650.73	31252.167
Valid N (listwise)	100				

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Oneway

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ANOVA

total_volume_usd

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	2.233E+17	2	1.117E+17	1.225	.298
Within Groups	8.842E+18	97	9.116E+16		
Total	9.066E+18	99			

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Post Hoc Tests

Multiple Comparisons

Dependent Variable: total_volume_usd

Tukey HSD

(I) PriceTirerNum	(J) PriceTirerNum	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
High	Low	-1920713.516	73780307.59	1.000	-177534161	173692734.4
	Medium	99542145.79	73780307.59	.372	-76071302.15	275155593.7
Low	High	1920713.516	73780307.59	1.000	-173692734	177534161.5
	Medium	101462859.3	74328867.61	.363	-75456282.81	278382001.4
Medium	High	-99542145.8	73780307.59	.372	-275155594	76071302.15
	Low	-101462859	74328867.61	.363	-278382001	75456282.81

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Homogeneous Subsets

total_volume_usd

Tukey HSD[a,b]

PriceTirerNum	N	Subset for alpha = 0.05 1
Medium	33	72710261.24
High	34	172252407.0
Low	33	174173120.5
Sig.		.360

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 33.327.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

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Correlations

Correlations

		floor_usd	total_volume_usd	owners_count
floor_usd	Pearson Correlation	1	.043	-.082
	Sig. (2-tailed)		.668	.417
	N	100	100	100
total_volume_usd	Pearson Correlation	.043	1	.046
	Sig. (2-tailed)	.668		.650
	N	100	100	100
owners_count	Pearson Correlation	-.082	.046	1
	Sig. (2-tailed)	.417	.650	
	N	100	100	100

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Regression

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	daily_volume_usd, owners_count, total_volume_usd[b]	.	Enter

a. Dependent Variable: floor_usd

b. All requested variables entered.

[redacted]

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.126[a]	.016	-.015	88746.406

a. Predictors: (Constant), daily_volume_usd, owners_count, total_volume_usd

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Initial Cluster Centers

	Cluster		
	1	2	3
Zscore(floor_usd)	-.28978	6.71619	-.28939
Zscore (total_volume_usd)	8.57362	-.25267	-.33254
Zscore (daily_volume_usd)	.52018	-.12537	-.43416
Zscore(owners_count)	-.23479	-.33603	6.69862

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Iteration History[a]

	Change in Cluster Centers		
Iteration	1	2	3
1	4.677	6.366	6.503
2	1.600	.592	.149
3	.666	.364	.037
4	.000	.280	.029
5	.000	.769	.054
6	.000	3.304	.091
7	.000	.000	.000

a. Convergence achieved due to no or small change in cluster centers. The maximum absolute coordinate change for any center is .000. The current iteration is 7. The minimum distance between initial centers is 9.933.

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Final Cluster Centers

	Cluster		
	1	2	3
Zscore(floor_usd)	.60267	5.18403	-.19315
Zscore (total_volume_usd)	3.33008	-.18554	-.13724
Zscore (daily_volume_usd)	3.66768	-.24038	-.15000
Zscore(owners_count)	-.13153	-.33331	.01641

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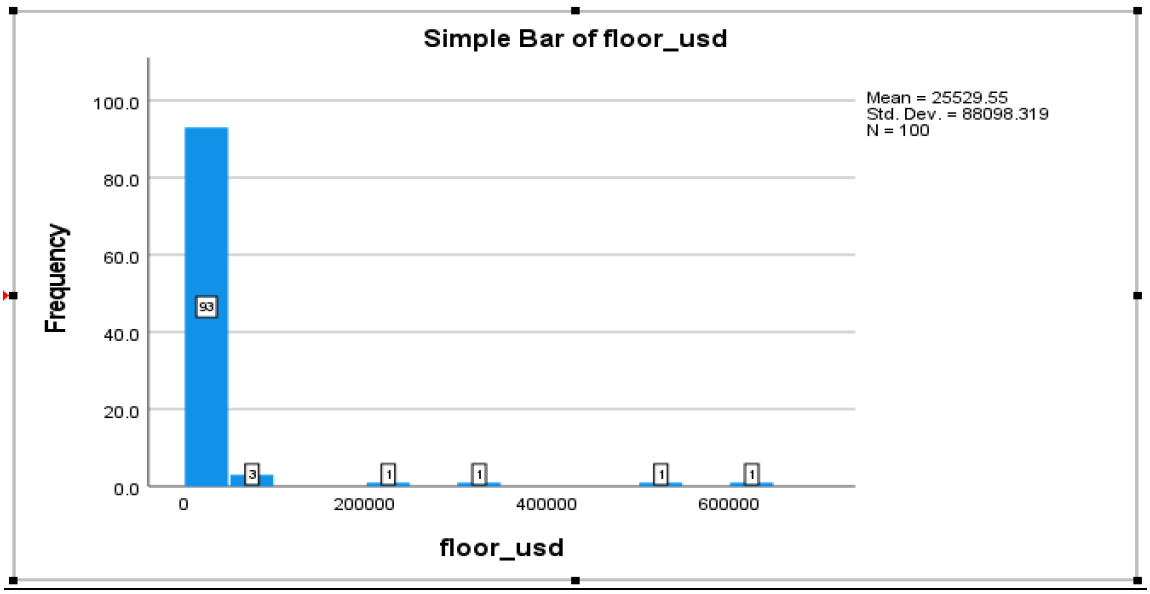

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Number of Cases in each Cluster

Cluster	1	4.000
	2	3.000
	3	93.000
Valid		100.000
Missing		.000

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ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.228E+10	3	4093766419	.520	.670[b]
	Residual	7.561E+11	96	7875924596		
	Total	7.684E+11	99			

a. Dependent Variable: floor_usd

b. Predictors: (Constant), daily_volume_usd, owners_count, total_volume_usd

[redacted]

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	23541.229	10665.601		2.207	.030
	total_volume_usd	6.117E-6	.000	.021	.198	.844
	owners_count	-.220	.286	-.078	-.769	.444
	daily_volume_usd	.005	.006	.088	.827	.410

a. Dependent Variable: floor_usd

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